1 Cheat sheet

Page 1 of 2: set up EVA. EVA is a local character you can talk to in Unity. Choose English or German before speaking. Installation needs Internet; the downloaded speech and answer models run on your PC.

What you need

Software	Windows x64, Python **3.11 x64**, Unity **6000.3.12f1**, Unity Hub, Git with Git LFS, and Ollama.
Graphics card	An NVIDIA card with CUDA and FP16 support and a suitable driver. This version cannot run on the CPU alone.
Previous PCs	Desktop: RTX 5070 Ti, 16 GB graphics memory, Ryzen 7 9700X, 32 GB RAM. Laptop: RTX 5070 Laptop GPU with about 8 GB graphics memory.
Suggested PC	Estimate: a recent 6-core CPU, 16 GB RAM (32 GB recommended), an NVIDIA RTX GPU with 8 GB graphics memory (12–16 GB recommended), and an SSD.
Space and limits	Start with at least 12 GB free for backend setup. Unity, assets and the import cache need extra space. These are estimates, not measured minimum requirements. Run the project checks on the actual PC.

1. Copy the project and install the software

Keep `Assets`, `Packages`, `ProjectSettings`, `Backend` and the project instructions. Create a new `Backend/.venv` and let Unity create a new `Library`. Keep existing models and voices. A copied Python environment may cause *No Python at* / exit code 103.

Open PowerShell in **Backend**:

```
Set-ExecutionPolicy -Scope Process Bypass
.\hardware_report.ps1
# Read hardware-report.json. Check the GPU first.
.\setup_backend.ps1 -RecreateEnvironment -InstallOllama
```

Listing 1: Check the PC and install the backend

The NVIDIA setup creates a Python environment, installs the pinned packages and checks or downloads Whisper Small, Llama 3.2 3B and all four Piper voices. It supplies CUDA runtime libraries; a full CUDA developer toolkit is normally unnecessary. Appendix A of the full report lists the package versions.

2. Check the backend

```
.\.venv\Scripts\python.exe verify_backend.py
```

Listing 2: Run the setup check

The last line must say:

```
[EVA Check] SUCCESS: The local GPU backend is ready.
```

For missing Microsoft runtime DLLs, install the Visual C++ x64 Redistributable. See the ONNX Runtime requirements.

3. Open Unity

Use Unity **6000.3.12f1** (recommended). Newer versions may work but are untested. Open `Assets/_Project/Scenes/Main.unity`, wait for import and fix compiler errors before Play. This is also the configured build scene. The launcher starts or reuses Ollama and Python. Normal installations need no manual path overrides or extra PowerShell server windows.

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Choose your EVA after loading

Press Play and wait for loading. Input stays blocked until the backend sends its readiness message. Then use the selection menu:

1. **01 CHARACTER:** choose one of six designs. Only one EVA is visible.
2. **02 BEHAVIOUR:** choose Neutral, Friendly or Expressive body animations.
3. **03 LANGUAGE:** choose English or Deutsch. This fixes recognition, answer text and speech to that language. There is no automatic language switching.
4. **04 VOICE:** click a name to select it, independently of the design. The small play icon previews that voice without selecting it; click again to stop.

Select **Start conversation**. Confirmed choices are saved locally, with a separate voice choice per language. A preview stops before conversation starts.

Voice	English	German
Female	Kristin, 22,050 Hz	Ramona, 16,000 Hz
Male	Mike, 22,050 Hz	Thorsten-high, 22,050 Hz

Hz is the audio sample rate: samples of the sound wave per second. For example, 16,000 Hz means 16,000 samples per second. Unity uses the supplied rate so speech plays at the correct speed and pitch.

Desktop controls

Before Play, choose EVA > Controls > Desktop (WASD + Mouse). After starting conversation, click empty Game-view space to capture the mouse. **WASD**: move; **mouse**: look; **Shift**: move faster; **Esc**: release the mouse. Hold **B** to speak, then release to send. A new recording interrupts the old answer. Press **M** to reopen settings when EVA is idle. Change language there before speaking another language.

Text boxes start hidden. Press **T** to show/hide the recognized question and EVA's answer. Both reveal words as you speak or EVA generates a reply. With the mouse released, use their move and resize handles.

Meta Quest 3 controls

Start Quest Link or Air Link first; choose EVA > Controls > VR (Meta Quest). In the menu, use the **right stick** and **A** to navigate/confirm. Right **B** opens settings while idle or closes the menu with Start/Continue. During conversation, left stick moves; right stick snap-turns by 30 degrees. Hold right **A** to record and release to send; release the menu confirmation press first. Left **X** shows/hides text; left **Y** repositions the surfaces. Menu and text sit near EVA, about 2–3 m away, and stay in the scene when you turn your head. In VR, text starts visible. Adjust size/distance in `Assets/_Project/Resources/EVAControlSettings.asset`, under VR panels (metres). Unity and Python run on the PC, not the headset.

Quick checks and fixes

Try both menu languages and both voices in each. Listen to previews and complete answers; check lip-sync, styles, gaze, text boxes and interrupting speech. Repeat with the real microphone and headset before a demo.

CUDA/DLL error: rerun setup with the project Python; do not mix CPU and GPU ONNX Runtime packages.

No connection: read `Backend/backend-runtime.log` and the Console. Stop Play, fix the cause and restart. For exit 103, recreate the environment.

VR closes: start Quest Link first and fully restart Unity.

Missing speech: long answers can exceed the scene's 60-second audio queue.